Arunabha Majumder

Mechanical | Mechatronics Engineer

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EDUCATION

PhD, Mechatronics and Robotics
Aalborg University, Denmark
2023 - Present (thesis expected March)
Study abroad: National University of Singapore (NUS), Sep - Nov 2025

MTech, Mechatronics
AcSIR - CSIR-CMERI, Durgapur, India
2020 - 2022

BTech, Mechanical Engineering
Siddaganga Institute of Technology (VTU), India
2015 - 2019

Diploma in Fine Arts (Painting)
Bangiya Sangeet Parishad, India

SKILLS

Design: mechanical product development, actuators and robotic mechanisms, Design for Assembly

Analysis: stiffness and kinematic modelling, stiffness analysis in SolidWorks, design optimisation with a genetic algorithm

Testing: prototype building, sensor integration, custom test rigs, test data processing

Visuals: hand sketching of concepts, technical illustrations for papers, posters and demos

TOOLS

CAD: SolidWorks (CSWA), 3D printing (Bambu Lab, Ultimaker Cura)

Software: MATLAB/Simulink, Arduino, LaTeX, Python and LabVIEW (basic)

Lab: Qualisys motion capture and data acquisition, Forsentec force/torque sensors, pneumatics

Graphics: Adobe Illustrator, Photoshop, Procreate

AWARDS

- Best Research Paper Award, 9th IFToMM ISRM (2026)
- Certified SolidWorks Associate (CSWA)
- Best Major Project Award (2019)
- TATA Technologies Innovation Grant, INR 100,000 (2019)

LANGUAGES

English (C1), Bengali (native), Hindi (spoken)

INTERESTS

Painting and illustration; built a Go-Kart (2017) and an Efficycle hybrid vehicle (2018) with student teams

REFERENCES

Available upon request.

PROFILE

Mechanical and mechatronics engineer who takes robotic hardware from a hand sketch to CAD, prototypes and testing with real people. I design actuators and exoskeleton mechanisms, integrate motors and sensors, and check designs against measurements. I explain ideas clearly with sketches, reports and talks.

EXPERIENCE

Research Engineer, Shoulder Exoskeleton (PhD project VIEXO)
Aalborg University, Denmark | *2023 - Present*

- Designed a hybrid shoulder exoskeleton where a parallel spring carries most of the arm load and a 6 Nm motor adds the rest; cut energy use by more than 25% (Best Paper Award, IFToMM ISRM 2026).
- Took the design from hand sketches to SolidWorks models and drawings, optimised it with a genetic algorithm, and measured the stiffness it provides on a physical test setup; 3D printed and rebuilt parts after each test.
- Integrated the motor, encoder and force sensors into the frame and built custom test rigs.
- Worked in a team of 3 on an Assist-as-Needed controller (my part: equations such as the Fourier fit, and defining the GUI behaviour); muscle effort dropped by more than 15%.
- Collected data from 12 participants (EMG with the Qualisys data acquisition system, and motion capture).
- Estimated tip force on a 5-bar parallel robot from its variable-stiffness actuators (within 10% RMS error vs a force sensor), removing the need for an extra sensor; mapped actuator stiffness to workspace stiffness.
- Implemented an Arduino-based trigger system to synchronise robot motion with Qualisys motion capture for arm impedance tests (6 participants).
- Taught university courses and supervised Bachelor's students (2023 - 2025).

MTech Researcher, Pneumatic Artificial Muscles
CSIR-CMERI (National Laboratory), Durgapur, India | *2020 - 2022*

- Fabricated and assembled pneumatic muscles to an existing lab design and collected test data on the lab's setup.
- Designed the outer PID loop of a neural network-based gain-scheduled controller in MATLAB/Simulink and implemented it in LabVIEW to drive the pneumatic muscle (0.3 - 0.78% position error, IEEE CONECCT 2022).

Bachelor's Capstone, Upper-Limb Rehabilitation Exoskeleton
Siddaganga Institute of Technology, India | *2018 - 2019*

- Designed and built an upper-limb rehabilitation exoskeleton for a paralysed arm, my first exoskeleton.
- Led CAD design, machining, assembly and user testing; Best Major Project Award and TATA Technologies grant.

PUBLICATIONS

- Majumder, A., Wagner, J.W., Zhu, Y., Oliveira, A.S., Bai, S. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." ISRM 2026, Springer. **Best Research Paper Award.**
- Majumder, A., Sarkar, D., Chakraborty, S., Singh, A., Roy, S.S., Arora, A. "Neural Network-Based Gain Scheduled Position Control of a Pneumatic Artificial Muscle." IEEE CONECCT 2022.